



King Mongkut's University of Technology Thonburi

Final Exam
CPE334 Operating Systems

Instructions

- This 11-page exam contains 6 questions for a total of 100 points.
- The answers must be written in this exam paper. Please read the instruction carefully.
- A calculator and a paper dictionary are allowed.
- A single A4-size note make be taken into the examination room (Write on both sides).

This examination is designed by Assoc.Prof.Dr. Tiranee Achalakul

Name _____ ID _____

Question 1	
Question 2	
Question 3	
Question 4	
Question 5	
Question 6	

Question 1: Multiple choices (10 Points)

Students get 1 point for each correct answer and -0.5 point for each wrong answer.

1. With DMA (Direct Memory Access):
 - a. The processor can read or write directly to a device.
 - b. The kernel can read or write directly to a process' memory without intermediate buffers.
 - c. A process can read or write to kernel memory without intermediate buffers.
 - ☒ d. The device can read or write directly to the system's memory.
2. To implement a user-level threads package, the operating system may need to provide:
 - ☒ a. Non-blocking system calls
 - b. Kernel threads
 - c. An exec mechanism
 - d. Direct memory access

User-level threads requires non-blocking systems call i.e., a multithreaded kernel. Otherwise, entire process will be blocked in the kernel, even if there are runnable threads left in the processes. For example, if one thread causes a page fault, the process blocks.
3. A compare-and-swap instruction allows you to:
 - a. Modify a memory location only if its contents match a given value.
 - ☒ b. Exchange the contents of two memory locations if their values are different.
 - c. Exchange the contents of two memory locations if a lock is not set.
 - d. Exchange the contents of two memory locations if a lock is set.
4. Starvation is the case when a thread:
 - a. Loops continuously until it runs out of memory.
 - ☒ b. is never scheduled to run.
 - c. Can never acquire a lock on a critical section.
 - d. Cannot create a child process or thread
5. A thread that is blocked on a semaphore is awakened when another thread:
 - a. Tries to decrement a semaphore's value below 0.
 - ☒ b. Tries to increment the semaphore.
 - c. Causes the semaphore's value to reach a specific number.
 - d. Tries to block on the same semaphore
6. Condition variables support these operations:
 - ☒ a. Wait / notify
 - b. Read-and-increment / wait-for-value
 - c. Increment / decrement-and-wait
 - d. Set-value / wait-for-value
7. Which statement is true:
 - ☒ a. Using mutual exclusion ensures that a system avoids deadlock.
 - b. The value of a semaphore can never be negative.
 - c. A Lock built with Test-and-Set can guarantee that a waiting thread will eventually enter the critical section.

d. Two phase locking use 'sleep' instead of 'spin' in its implementation

8. Which of the following is the allocation method of a disk space?

- a. Contiguous allocation
- b. Linked allocation
- c. Indexed allocation
- ☒ d. All of the above

9. At a particular time of computation, the value of a counting semaphore is 7. Then 20 'P' operations and 15 'V' operations were completed on this semaphore. The resulting value of the semaphore is:

- a. 42
- ☒ b. 2
- c. 7
- d. 12

$$\begin{array}{l} P \rightarrow \text{wait} - 20 \\ V \rightarrow \text{post} + 15 \end{array} \quad \left. \vphantom{\begin{array}{l} P \rightarrow \text{wait} - 20 \\ V \rightarrow \text{post} + 15 \end{array}} \right\} 7 - 5 = 2$$

10. The _____ present a uniform device-access interface to the I/O subsystem, much as system calls provide a standard interface between the application and the operating system.

- a. devices
- b. buses
- ☒ c. device drivers
- d. I/O systems

မာဝေခေအ်

Question 2: Short answer questions. Use only the space provided to write your answers. (10 points)

1. (2 points) Exceptions are events that stop normal execution, switch the execution mode into kernel mode, and begin execution at special locations within the kernel. Name two different types of exceptions and give an example of each type.

- divide by zero

- invalid memory access

2. (4 points) For each of the following thread state transitions, say whether the transition is legal. If so, explain how the transition occurs. If not, say why.

a) Change from thread state BLOCKED to thread state RUNNING

ไม่

}

มี semaphore

b) Change from thread state RUNNING to thread state BLOCKED

ใช่

มี semaphore
ให้ไปเอา

3. (2 points) Given that we can create user-level code to control access to critical sections, why is it important for an operating system to provide synchronization facilities such as semaphores in the kernel?

software จะเกิดปัญหา spin-wait

โดยต้องให้คำสั่งของ kernel มาช่วยทำให้ sleep หรือ mutex spin-wait

Question is about offering sync services via the kernel than via user-level code.

To avoid busy waiting: the waiting thread can go to sleep

→ creates better cpu utilization; avoids priority inversion (if an issue)

4. (2 points) Name at least three things that affect the speed of the disk access time.

transfer rate

rotation time

seek time

Question 3: (20 points)

3.1 (8 points) The methods 'Lock' (wait) and 'Unlock' (signal) must be atomic to ensure a correct implementation of mutual exclusion. Give an example scenario to show that if lock() is not atomic then mutual exclusion cannot be maintained. The scenario must be explained with both Pseudo-code and plain language (English or Thai)

implement lock ด้วยวิธี test and set

```
lock(flag) {  
    while(flag == 1)  
        flag = 1  
}
```

ตอนเริ่ม flag=1

สมมุติ มี process 1 เข้ามา lock ก็จะเห็นว่า flag=1 จึง

process 1: while(flag==1)

แต่ flag=0 อยู่จึงจะไปทำงานต่อ คือ set ค่า flag=1 แทน

ก่อนจะ set ค่าไป context switch ไป process 2

process 2 ก็จะเห็นค่า flag

process 2: while(flag==1)

ซึ่ง flag=0 อยู่จึงไปทำงานต่อ คือ set ค่า flag

process 2: flag=1

และมีการ context switch กลับไป process 1 ให้ทำงานต่อ

process 1: flag=1

ทำให้ไม่มีใคร spin-wait อยู่ใน lock และ 2 process ก็จะ
เข้าไปทำงานพร้อมๆ กัน

3.2 (8 points) Compare the two concepts of 'conditional variable' and 'semaphore' by given definition for both terms and explicitly state the differences between the two.

condition variable คือ การทำ synchronization
จัดการลำดับการทำงานของ thread

semaphore จะทำทั้ง synchronization และ lock ให้เป็น
mutual exclusion

3.3 (4 points) Give a scenario where choosing a large filesystem block size might be a benefit; give an example where it might be of disadvantage.

ข้อดี
↳ 1 block เก็บข้อมูลได้เยอะ
↳ access 1 block ได้ข้อมูลเยอะ
ข้อเสีย
↳ internal fragmentation

Question 4: (20 points)

Imagine a number of concurrent list operations, including Inserts and simple lookups.

1. Inserts change the state of the list, and thus only a single writer can acquire the lock to enter the critical section.
2. Lookups simply read the list; as long as we can guarantee that no insert is on-going, we can allow many lookups to proceed concurrently.

Write a pseudo-code to use the semaphore concept to implement a special type of lock to support this type of operation (known as *reader-writer lock*).

Hint: If some thread wants to update the list, it should call the new pair of synchronization operations:

- `rwlock_acquire_writelock()`, to acquire a write lock
- `Rwlock_release_writelock()`, to release it.

If some thread wants to read from the list, it should call:

- `rwlock_acquire_readlock()`, to acquire a read lock
- `Rwlock_release_readlock()`, to release it.

Now use semaphore to write a pseudo-code for these four functions.

```
sem_t empty, full, mutex;  
sem_init(&empty, 0, 1); // if empty = 1 → list is empty  
sem_init(&full, 0, 0); // if full = 0 → list is full  
sem_init(&mutex, 0, 1); // if mutex = 1 → lock is on  
  
void *rwlock_acquire_writelock() {  
    sem_wait(&empty);  
    sem_wait(&mutex);  
}  
  
void *Rwlock_release_writelock() {  
    sem_post(&full);  
    sem_post(&mutex);  
}  
  
void *rwlock_acquire_readlock() {  
    sem_wait(&full);  
    sem_wait(&mutex);  
}  
  
void *Rwlock_release_readlock() {  
    sem_post(&empty);  
    sem_post(&mutex);  
}
```

Question 5: (15 points)

5.1 (5 points) Processor and I/O Controller can work together in two different ways. First, the processor uses generic I/O commands with Memory-mapped I/O. Second, processor sends special I/O commands directly to each I/O controller. Explain the mechanism of both methods and discuss their differences.

แบบ Memory-mapped I/O

↳ device register ถูก mapped ลง memory space
เพื่อใช้คำสั่ง store/load ปกติ

แบบ special I/O command

↳ ใช้คำสั่งพิเศษ read/write ไปที่ device register โดยตรง

5.2 (5 points) Compare I/O based on polling with interrupt-driven I/O. In what situation would you favor one technique over the other?

polling เหมาะกับ device ที่ transfer data เร็ว,
ไม่มีความจำเป็นเวลา interrupt

interrupt เหมาะกับ HCI (human computer interface)

5.3 (5 points) When a file is removed, the blocks it occupies are simply placed back onto the free list. Data can still be accessed with the correct tools, which may be problematic. How would you overcome such a problem? Describe your idea.

เวลา delete file คือจะลบชื่อfileกับinode ออกจาก directory
คือ จะมองไม่เห็น แต่จริงๆ file ยังอยู่
แก้ด้วยการทำ defragment หรือใช้ format เลย!

Question 6: (25 points)

6.1 (5 points) Explain briefly how to create your own system call step by step

1. สร้าง file code ของเรา
2. เพิ่ม file ลงใน kernel's makefile
3. เพิ่ม file ลง system call table
4. เพิ่ม system call ในไฟล์ syscall.h
5. compile kernel
6. update kernel

6.2 (5 points) Write a code to test a system call if the number of system calls is 323 and the function of a system call is show below

```
asm linkage long sys_mod (long a, long b)
{
    long answer;
    answer = a + b;
    return answer;
}
```

```
int main()
{
    long no = syscall(323);
    printf("sys_mod return %ld\n", no);
    return 0;
}
```

5. (15 points) Explain how you can achieve your task in assignment 4 (CPU, Memory, Threads, File system). The explanation must include source file name that you have edited.

9x10⁹ 2¹⁵
v v

